Parity, conservation of, 40–43
  failure of, 42–43
Past (time-like interval), 99–100
Philosophers, and relativity, 73–77
Photons, 108–109, 135–136
Physics, laws of. *See also* specific laws, e.g. Conservation laws
  change of scale, effect of, 26–28, 39
  invariance of, 9, 16
  reversibility in time of, 28–29
  symmetry of, 2–3, 77
  vector notation, 16
Planck's constant, 108
Poincaré, Henri, xii, xiii, xv, 54, 58, 73
Polar vectors, 35–38
Positrons, 43–44, 71
Protons, 44, 46

Quantum mechanics, ix, xv, 29, 30

Reflection symmetry, xiv, 30–35
  antimatter and, 45
  identification of right and left, 38–39
  polar and axial vectors, 35–38
Relativity, *See also* Lorentz transformation
  addition of velocities in, 80–82
  energy, relativistic, xvi, 88–91
  General Theory of, xv, 50
  of mass, xvi, 49–50, 66–68, 83–88
  Michelson and Morley experiment, 54–59, 60, 63
  momentum, law of conservation of, 107
  philosophers and, 73–77
  principle of, 49–53, 73, 75
  relativistic dynamics, 66–68
  simultaneity, 63–64
  Special Theory of, ix, xv, 49–71
  time, transformation of, 59–62
Right-handedness, 31–39
Rotation
  Lorentz transformation, 65
  space-time measurements, 93–94
  symmetry and, xiv, 5–9, 16, 25, 28, 29
  uniform, 76

Sands, Matthew, xi, xxviii
Scalar, 10, 19
Scalar function, 19
Scalar product, 18–21, 108
Scale, changes in, 26–28, 39
Schrödinger equation, 33–34
Schwinger, Julian, xxii
Seconds, measure of distance by, 98
Simultaneity, 63–64, 101
  failure at a distance, 64
Sound waves, speed of, 52
Space
  curved. *See* Curved space
  reflection in, 30–35
  rotation in, 25, 28, 29
  space-time compared, 99
  symmetry in, 24–29
Space-like interval, 99–100
Space-time
  curvature of, xiv–xv, 111, 137–141